

Figure 1: Data hazard between i0 and i1

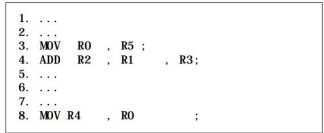


Figure 4: Input set of instructions

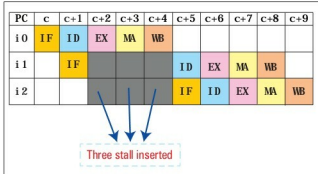


Figure 2: Inserting stalls to overcome data hazard

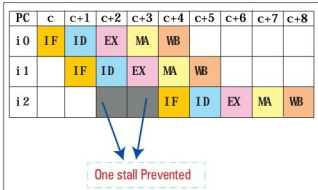


Figure 3: One stall prevented by instruction rescheduling

its source operand in the *ID* stage itself. Thus, the *MUL* instruction may pick up the wrong value of *R1*, since the *WB* of the previous instruction has not happened by the time the *ID* of the current instruction happens. To overcome this hazard, the processor inserts three stalls in the pipeline, as shown in Figure 2.

Now, if you notice the input set of instructions,

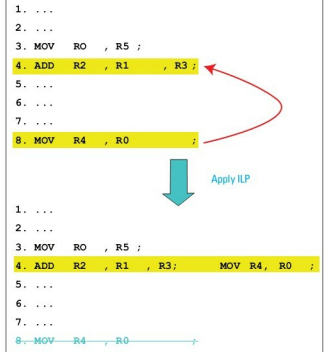


Figure 5: Applying ILP on the input set of instructions

Instruction *i2* is an independent instruction (it does not have dependency with any other input instruction). Thus, we may reschedule the instructions as follows:

```
ADD R1, R2, R3 //Instruction i0
MOV R5, R6    //Instruction i2 -> rescheduled
MUL R4, R1, R2 //Instruction i1
```

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